

MAT 344 HW01

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Problem 1 (p.6#2). Show that the complex number system can be thought of as the field of all polynomials with real coefficients modulo the irreducible polynomial $x^2 + 1$.

Solution. Let $\varphi: \mathbb{R}[x] \rightarrow \mathbb{C}$ be the homomorphism determined by

$$\varphi(a) = a \text{ for } a \in \mathbb{R}, \quad \varphi(x) = i.$$

The kernel of this homomorphism contains the polynomial $x^2 + 1$ because $\varphi(x^2 + 1) = i^2 + 1 = 0$ and does not contain any nonzero polynomials of smaller degree because $\varphi(a + bx) = a + bi$ is zero if and only if $a = b = 0$. Since $\mathbb{R}[x]$ is a principal ideal domain, the kernel of φ is generated by $x^2 + 1$ (a nonzero polynomial in the ideal with minimal degree). By the first isomorphism theorem for rings, φ descends to an isomorphism between the fields $\mathbb{R}[x]/(x^2 + 1)$ and $\mathbb{C}$. $\square$

Problem 2 (p.11#1). Prove that

$$\left| \frac{a - b}{1 - \bar{a}b} \right| < 1$$

if $|a| < 1$ and $|b| < 1$.

Solution. It suffices to prove that $|a - b|^2 < |1 - \bar{a}b|^2$. The magnitudes can be rewritten as

$$\begin{aligned} |a - b|^2 &= (a - b)(\bar{a} - \bar{b}) = a\bar{a} + b\bar{b} - \bar{a}b - a\bar{b} \\ |1 - \bar{a}b|^2 &= (1 - \bar{a}b)(1 - a\bar{b}) = 1 + a\bar{a}b\bar{b} - \bar{a}b - a\bar{b}. \end{aligned}$$

Using $|a|, |b| < 1$,

$$1 + a\bar{a}b\bar{b} - a\bar{a} - b\bar{b} = (1 - a\bar{a})(1 - b\bar{b}) = \underbrace{(1 - |a|^2)}_{>0} \underbrace{(1 - |b|^2)}_{>0} > 0 \implies a\bar{a} + b\bar{b} < 1 + a\bar{a}b\bar{b}.$$

Thus,

$$|a - b|^2 = a\bar{a} + b\bar{b} - \bar{a}b - a\bar{b} < 1 + a\bar{a}b\bar{b} - \bar{a}b - a\bar{b} = |1 - \bar{a}b|^2$$

as desired. $\square$

Problem 3 (p.20#1). Show that z and z' correspond to diametrically opposite points on the Riemann sphere if and only if $zz' = -1$.

Solution. Let N be the north pole of the Riemann sphere and Z and Z' be the points on the sphere corresponding to z and z' , i.e. Z is the second intersection of line Nz with the sphere and Z' is the second intersection of line Nz' with the sphere. Letting O be the origin of the complex plane (i.e. 0), we have that $\angle NOz = \angle NOz' = 90^\circ$.

If ZZ' is a diameter of the sphere (and Z, Z' are not N , so that neither z nor z' is ∞), then $\angle zNz' = \angle ZNZ' = 90^\circ$ and O lies on segment zz' . The latter statement is true because the circle passing through Z, N , and Z' , which contains the south pole, gets mapped via stereographic projection to a line in the plane containing z, z' , and the image O of the south pole (to be precise, O lies on *segment* zz' because the south pole lies on the arc connecting Z and Z' that doesn't contain N). This is all to say that O, N, z , and z' are coplanar. Also, $\triangle zON \sim \triangle z'Nz'$ because both triangles have a right angle and share the

angle $\angle NzO = \angle Nzz'$, and $\triangle z'ON \sim \triangle z'Nz$ by an identical argument. This means we have the ratio of distances $zO/ON = NO/Oz'$, and since $ON = 1$,

$$|z| = zO = \frac{1}{Oz'} = \frac{1}{|z'|} \iff |zz'| = 1.$$

The fact that O lies on segment zz' additionally means that $\frac{z}{z'} = cz\overline{z'}$ is a negative real number (where $c = \frac{1}{|z'|^2} > 0$), so $zz' = -1$.

Suppose conversely that $zz' = -1$. Since this product is a negative real number, O still lies on segment zz' , meaning Z and Z' are concyclic with the north and south poles of the sphere. Since the product has magnitude 1, $Oz/ON = NO/Oz'$, which together with $\angle NOz = \angle NOz' = 90^\circ$ implies that $\triangle zON \sim \triangle NOz'$ are similar right triangles. Then,

$$\angle zNz' = \angle zNO + \angle ONz' = \angle zNO + \angle OzN = 180^\circ - \angle NOz = 90^\circ,$$

where the first equality holds because O is on segment zz' , so $\angle zNz' = \angle ZNZ' = 90^\circ$. Together with the fact that Z and Z' lie on a great circle of the sphere (the great circle through N and its antipode, specifically), we deduce that ZZ' is a diameter of the sphere. $\square$

Problem 4 (p.28#4). Show that an analytic function cannot have a constant absolute value without reducing to a constant.

Solution. Let $f = f_1 + if_2$ for real-valued functions f_1, f_2 . We are given that $f_1^2 + f_2^2$ is constant, so

$$\begin{aligned} \frac{\partial}{\partial x}(f_1^2 + f_2^2) &= 2f_1 \cdot \frac{\partial f_1}{\partial x} + 2f_2 \cdot \frac{\partial f_2}{\partial x} = 0 \\ \frac{\partial}{\partial y}(f_1^2 + f_2^2) &= 2f_1 \cdot \frac{\partial f_1}{\partial y} + 2f_2 \cdot \frac{\partial f_2}{\partial y} = 0. \end{aligned}$$

Let $z \in \mathbb{C}$ be arbitrary; we will show that the above two equations imply $f'(z) = 0$, which, since $\mathbb{C}$ is connected, implies f is constant. If both $f_1(z)$ and $f_2(z)$ are zero, then $|f(z)| = 0$ everywhere, so f is constantly zero. If exactly one of $f_1(z)$ and $f_2(z)$ is zero, say $f_1(z) = 0$, then we may divide by $f_2(z)$ to get

$$\frac{\partial f_2}{\partial x}(z) = \frac{\partial f_2}{\partial y}(z) = 0.$$

By the Cauchy-Riemann equations, all the partial derivatives of f are zero, since $\frac{\partial f_2}{\partial x} = -\frac{\partial f_1}{\partial y}$ and $\frac{\partial f_2}{\partial y} = \frac{\partial f_1}{\partial x}$. This implies $f'(z) = 0$.

Finally, suppose that $f_1(z)$ and $f_2(z)$ are both nonzero. Rearranging and multiplying the equations yields

$$f_1 f_2 \cdot \frac{\partial f_1}{\partial x} \cdot \frac{\partial f_2}{\partial y} = f_1 f_2 \cdot \frac{\partial f_1}{\partial y} \cdot \frac{\partial f_2}{\partial x} \implies \frac{\partial f_1}{\partial x} \cdot \frac{\partial f_2}{\partial y} = \frac{\partial f_1}{\partial y} \cdot \frac{\partial f_2}{\partial x}.$$

By the Cauchy-Riemann equations, the left-hand side of the latter equation is equal to $(\frac{\partial f_1}{\partial x})^2 = (\frac{\partial f_2}{\partial y})^2 \geq 0$ and the right-hand side is equal to $-(\frac{\partial f_1}{\partial x})^2 = -(\frac{\partial f_2}{\partial y})^2 \leq 0$, so the LHS and RHS are equal to zero. This means that each of the partial derivatives of f is zero, hence $f'(z) = 0$. $\square$

Problem 5 (p.28#5). Prove rigorously that the functions $f(z)$ and $\overline{f}(\overline{z})$ are simultaneously analytic.

Solution. Let $f: U \rightarrow \mathbb{C}$ be an arbitrary function on an open subset $U \subseteq \mathbb{C}$. Then, write

$$f(x + iy) = f_1(x + iy) + if_2(x + iy)$$

for functions $f_1, f_2: U \rightarrow \mathbb{C}$ and real variables x, y . Letting $g(z) = \overline{f}(\overline{z}) = f_1(x - iy) - if_2(x - iy)$, we have

$$\frac{\partial}{\partial \overline{z}} g = \frac{1}{2} \left(\frac{\partial f_1}{\partial x} - \frac{\partial f_2}{\partial y} \right) - \frac{i}{2} \left(\frac{\partial f_1}{\partial y} + \frac{\partial f_2}{\partial x} \right) = \overline{\left(\frac{\partial}{\partial z} f \right)}.$$

This means the left-hand side is zero ($\overline{f}(\overline{z})$ is holomorphic) if and only if the right-hand side is zero (f is holomorphic). $\square$

Problem 6 (p.28#6). Prove that the functions $u(z)$ and $u(\bar{z})$ are simultaneously harmonic.

Solution. Let $v(z) = u(\bar{z})$ and $u(z) = u_1(z) + iu_2(z)$ for real-valued functions u_1, u_2 . Since conjugation preserves the real part, $\frac{\partial^2}{\partial^2 x} v = \frac{\partial^2}{\partial^2 x} u$, and

$$\frac{\partial^2}{\partial^2 y} v(z) = \frac{\partial}{\partial y} \left(-\frac{\partial u}{\partial y}(\bar{z}) \right) = \frac{\partial^2 u}{\partial^2 y}(\bar{z})$$

where the negation in the middle term comes from the negation of the imaginary part in $\bar{z}$. Thus $\Delta u = \Delta v$ everywhere, so u is harmonic if and only if v is. $\square$